ASCOMP 2026 POSTER

Min-Marginal Power Indices for Multi-Objective Feature Selection

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01 CORE IDEA & METHOD

From average importance to worst-case evidence.
MinPI detects feature additions that push utility below the quota.



THE WORST-CASE MARGINAL

$$v_\alpha(S) = 1[U(M_S) \geq \alpha\, U_{full}]$$

$$\mathbf{MinPI_i = \min [v_\alpha(S \cup \{i\}) - v_\alpha(S)]}$$

over $S \subseteq N \setminus \{i\}$; N = analyzed features

Validation-gated feature removal

H: features with an evaluated harmful coalition witness (-1).

$$\mathbf{U_{val}(N \setminus H) \geq U_{val}(N) - \varepsilon}$$

ε = 0.005 (classification); 0.01 (regression).

Pass: remove H. Fail: remove nothing. Models fit on training data.

SCAN ME

CORE INSIGHT

From harmful-coalition evidence to validation-gated feature removal.

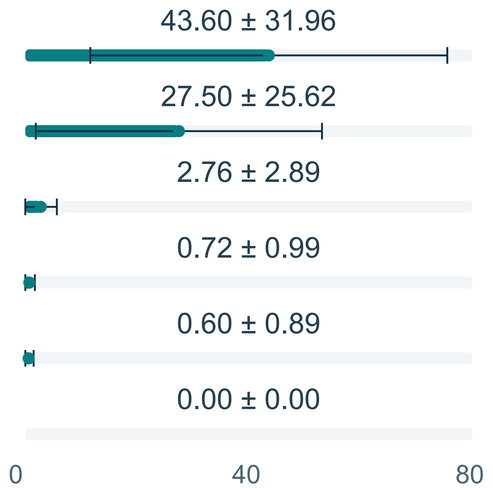
Scan to read the full paper
hlnks.co/63671101

02 EXPERIMENTS & RESULTS

6 datasets × 5 seeds | XGBoost | α = 0.95
64/16/20% train/validation/test; ≤100 analyzed features.

Five-seed mean ± SD; removal of analyzed features.
ΔU = held-out utility - full-model utility.

DATASET	REMOVED (%)	HELD-OUT ΔU
Madelon	43.60 ± 31.96	**+0.00469** ± 0.00497
Abalone	27.50 ± 25.62	**-0.00565** ± 0.01077
Credit-Card-Fraud	2.76 ± 2.89	**-0.00078** ± 0.00335
Spambase	0.72 ± 0.99	**-0.00003** ± 0.00007
Gisette	0.60 ± 0.89	**-0.00011** ± 0.00016
ISOLET	0.00 ± 0.00	**0.00000** ± 0.00000



12.53%
mean feature removal

-0.000313
mean held-out ΔU

NSGA-II: mean held-out ΔHV +0.01584; 800 evaluations/variant.
Madelon: +0.09515. Overall Holm p = 0.4375.

MADELON: MATCHED-COUNT REMOVAL

MinPI ΔU +0.0047 | Random ΔU -0.0732